

Program	B. Tech CSE: All Batches	Semester	I
Course	Physics for Computer Engineers	Course Code	PCE1001_5
Session	July-Dec 2026	Topic	Unit I: Lasers, Fibre Optics & Holography

Key Points and Formulae

- LASER is an acronym for Light Amplification by Stimulated Emission of Radiation. It is a device to produce an intense, monochromatic, collimated and highly coherent beam of light. The operation of a LASER depends upon the phenomenon of stimulated emission.
- Population inversion is a necessary criterion for laser action; if N_2 and N_1 are the number of atoms in excited and ground states respectively, then population inversion may be mathematically written as $N_2/N_1 > 1$. If E_2 and E_1 are the energies of the excited and ground states respectively, then $N_2/N_1 = \exp\left[-\left(\frac{E_2 - E_1}{KT}\right)\right]$, where K is the Boltzmann constant.
- Angular Spread: $d\theta = 1.22\lambda/D$, λ is the wavelength and D is the diameter of the front mirror.
- Areal spread $= (f \cdot d\theta)^2$, where f is the focal length and $d\theta$ is the angular spread.
- The ratio of stimulated and spontaneous emissions is given by $S_1 = \frac{1}{\exp(h\nu/KT)-1}$; ν is the frequency of laser light.

MULTIPLE-CHOICE QUESTIONS

1. He-Ne Laser is a
 - (a) 3-level Laser
 - (b) 4-level Laser
 - (c) 2-level Laser
 - (d) None of these
2. What color of light does the Ruby LASER emit?
 - (a) Blue
 - (b) Yellow
 - (c) White
 - (d) Red
3. Out of the following, which one is a two-level LASER?
 - (a) CO₂ LASER
 - (b) He-Ne LASER
 - (c) Ruby LASER
 - (d) Semiconductor LASER

4. The directionality of the LASER beam is measured by
- (a) Visibility of Interference fringes
 - (b) Size and aperture of the laser source
 - (c) the divergence angle of the beam with distance from the source
 - (d) Nature of the lasing medium

TUTORIAL SHEET

1. If level 1 and level 2 are separated by an energy $E_2 - E_1$ such that the corresponding transition frequency falls in the middle of the visible range, Calculate the ratio of the population of the two levels in thermal equilibrium at room temperature. **Ans: 1.357×10^{-38}**
2. Find the intensity of a laser beam of 100 mW power and having a diameter of 1.3 mm. Assume the intensity to be uniform. **Ans: 75.37 kW/m^2**
3. The populations of the lower and upper laser levels are:
- $$N_1 = 8 \times 10^{18} \text{ atoms/m}^3$$
- $$N_2 = 1.2 \times 10^{19} \text{ atoms/m}^3$$

Determine whether population inversion exists and calculate the population difference.

Ans: $4 \times 10^{18} \text{ atoms/m}^3$

4. Calculate the population ratio of two states in He- Ne Laser that produces light of wavelength 6000 Å at 300 K? **Ans: e^{-80}**
5. A laser emits radiation of wavelength 500 nm with a pulse energy of 2 mJ. Calculate the number of photons emitted. **Ans: 5.03×10^{15}**
6. Calculate the ratio of stimulated to spontaneous emission at a temperature of 300°C for Sodium D line. **Ans: 5.64×10^{-19}**
7. The first line of the principal series of sodium D-Line at 5890 Å corresponds to a transition from the first excited state to the ground state. What is the energy (in eV) of the first excited state? **Ans: 2.1063 eV**
8. Calculate the wavelength of laser emitted from an extrinsic semiconductor laser if the band gap is 0.02 eV. To which region of EM spectrum does it belong? **Ans: 618.08 nm, visible**
9. The ratio of population of two energy levels is 1.059×10^{-30} . Find the wavelength of light emitted by spontaneous emissions at 330K. **Ans: 632 nm**

10. The average output power of Laser source emitting a laser beam of wavelength 632.8 nm is 5mW. Find the number of photons emitted per second by the laser source.

Ans: $159.19 \times 10^{14}/s$

HOLOGRAPHY

Key Points and formula

- Holography was invented by English Physicist Denis Gabor in 1947.
- Light waves reflected from an object are recorded. This photograph record is called a hologram.
- Holography is a two-stage process: In the first stage, a hologram is recorded in the form of an interference pattern. In the second stage, the hologram acts as a diffraction grating for the reconstruction beam, and the image of the object is reconstructed from the hologram.
- Several applications like data storage, security, and holographic interferometry, etc. are few applications.

MULTIPLE-CHOICE QUESTIONS

1. Holography records ____ of light coming from an object.
(a) Intensities (b) Phases
(c) Both (a) and (b) (d) None of the above
2. Through Holography we can produce ____ dimensional image of objects
(a) One (b) Two
(c) Three (d) None of the above
3. Holography Produces ____
(a) Real images (b) Virtual
(c) Both (a) and (b) (d) None of the above
4. A recorded holographic plate contains information in the form of ____ Pattern
(a) Interference (b) Diffraction
(c) Both (a) and (b) (d) None of the above.
5. In holography, the interference pattern is produced from ____
(a) Object Beam (b) Reference beam
(c) Both (a) and (b) (d) None of the above

SUBJECTIVE QUESTIONS

1. Mention any two differences between a hologram and a photograph.
2. What is holography? Describe the construction and reconstruction methods of a hologram?
3. Elaborate on the principle of light used in holography.
4. Explain some engineering applications of Holography.
5. What is Holographic data storage? Explain some non-optical applications of holography.

FIBER OPTICS

Key Points and formula

- If n_1 be the refractive index of the core and n_2 , be the refractive index of the cladding, then for total internal reflection
 - $n_1 > n_2$, i.e., the refractive index of the core should be greater than the cladding
 - The angle of incidence on the core-cladding interface (w.r.t. vertical) should be greater than the critical angle.
- When $\theta = \theta_c$, where θ is the angle of incidence and θ_c is the critical angle, the ray traverses along with the interface (i.e. the angle of refraction is 90°).
- The light should travel from the core to the cladding.
- Fiber optics is a technology related to the transportation of optical energy (light energy) in guiding media, specifically glass fibers.
- Based on transmission properties and the structure, we can categorize optical fibers as single-mode fiber or multimode fiber. The typical diameter of the core of the single-mode fiber is $10\mu m$ and that of multimode fiber ranges from $50\mu m$ to $100\mu m$.

MULTIPLE-CHOICE QUESTIONS

1. The basic material for glass fiber is
 - (a) B_2O_3
 - (b) P_2O_5
 - (c) GeO_2
 - (d) SiO_2
2. Propagation of light through a fiber core depends upon the phenomenon of
 - (a) Interference
 - (b) Diffraction
 - (c) Polarization
 - (d) Total internal reflection
3. The numerical aperture of an optical fiber depends upon
 - (a) Core refractive index
 - (b) critical angle
 - (c) Both (a) and (b)
 - (d) none of these
4. The main energy losses in optical fibers consists of
 - (a) Material loss
 - (b) scattering loss
 - (c) Bending loss
 - (d) All of the above

5. The angle of acceptance is maximum for a fiber of

- (a) the critical angle is minimum (b) the critical angle is zero
(c) the critical angle is maximum (d) the critical angle is negative

TUTORIAL SHEET

1. A step-index fiber has a numerical aperture of 0.26, a core refractive index of 1.5, and a core diameter of 100 μm . Calculate the refractive index of the cladding. **Ans: 1.4772**
2. Calculate the numerical aperture of an optical fiber whose core and cladding are made of materials of refractive indices 1.6 and 1.5, respectively. **Ans: $NA = 0.56$, $\theta_o = 34.1^\circ$**
3. The numerical aperture of an optical fiber is 0.25 and the relative refractive index difference is 0.02. Determine the refractive indices of the core and cladding of a fiber. **Ans: 1.25, 1.225**
4. Calculate the numerical aperture and the acceptance angle of optical fiber from the following data: the refractive index of the core is 1.55 and the refractive index of the cladding is 1.50. **Ans: 0.391; 23°**
5. The refractive indices for the core and cladding for a step-index fiber are 1.52 and 1.41, respectively. Calculate i) critical angle ii) numerical aperture iii) the maximum incidence angle. **Ans: $\theta_c = 68.1^\circ$, $NA = 0.568$, $\theta_o = 34.6^\circ$**
6. A step index fibre has a core diameter of 0.05 mm and numerical aperture of 0.22 at a wavelength of 8500 $\text{\AA}$. What are the normalised frequency (V_n) and number of modes guided in the core? **Ans: Number of modes = 825**
7. How many modes can propagate in a step-index fibre with a core diameter as 40 μm , if the refractive indices of its core and cladding are 1.461 and 1.456, respectively, and the light of wavelength is 8500 $\text{\AA}$? **Ans: Number of modes = 159**
8. A signal of power 5 μW exists just inside the entrance of 0.1 km long fibre. Calculate the absorption coefficient of the fibre if the power inside the fibre be 1 μW . **Ans: $\alpha = 70 \text{ dB/km}$**
9. An optical fibre cable 3.0 km long is made up of three 1.0 km length spliced together. The losses due to each length and splice are respectively 5 dB and 1.0 dB. What would be out put power if the input power is 5 mW? **Ans: $P_o = 0.080 \text{ mW}$**
10. Find out the maximum core diameter of an optical fibre whose core and cladding have refractive indices as 1.460 and 1.457, respectively, and which supports only one mode at $1.25 \times 10^{-6} \text{ m}$ wavelength. **Ans: 10.22 μm**